

ADVANCED QUANTUM MECHANICS

Exercise 1 – 14.10.2025



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Operators, kets and bras, exponentials of operators and commutators

We will review some of the basic math to be used in the lectures.

Operators and expectation values

We consider a two-level system spanned by the orthonormal basis states $\mathcal{B} = \{|0\rangle, |1\rangle\}$ and an arbitrary normalized state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

(i) Compute the norm $\langle\psi|\psi\rangle$ and write the matrix representation of the projector $|\psi\rangle\langle\psi|$ in the basis $\mathcal{B}$.

(ii) For the operator

$$\hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

evaluate the expectation value and variance:

$$\langle\sigma_z\rangle_\psi = \langle\psi|\hat{\sigma}_z|\psi\rangle, \quad (\Delta\sigma_z)^2 = \langle\psi|\hat{\sigma}_z^2|\psi\rangle - \langle\sigma_z\rangle_\psi^2.$$

Interpret the results for the states $|0\rangle$, $|1\rangle$, and $(|0\rangle + |1\rangle)/\sqrt{2}$.

(iii) Verify explicitly the commutation relation $[\hat{\sigma}_x, \hat{\sigma}_y] = 2i\hat{\sigma}_z$ using matrix multiplication, and explain why this prevents the existence of simultaneous eigenstates of all three Pauli operators.

Reminder:

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

(iv) Insert the completeness relation $\sum_i |i\rangle\langle i| = \mathbb{1}_2$ into the expression for the expectation value and show that

$$\langle\psi|\hat{\sigma}_z|\psi\rangle = \sum_i |\langle i|\psi\rangle|^2 (\hat{\sigma}_z)_i,$$

interpreting the result as a probabilistic average over measurement outcomes.

Exponentials of operators

The exponential of an operator is defined through its power series:

$$e^{\hat{A}} = \sum_{n=0}^{\infty} \frac{1}{n!} \hat{A}^n = \mathbb{1} + \hat{A} + \frac{1}{2!} \hat{A}\hat{A} + \frac{1}{3!} \hat{A}\hat{A}\hat{A} + \dots$$

- (i) For a projection operator $\hat{P} = |0\rangle\langle 0|$ with $\hat{P}^2 = \hat{P}$, show that

$$e^{\lambda \hat{P}} = \mathbb{1} + (e^\lambda - 1)\hat{P}.$$

- (ii) Compute the following exponential

$$e^{i\theta \hat{\sigma}_x} = \cos \theta \mathbb{1}_2 + i \sin \theta \hat{\sigma}_x.$$

Show that this is a unitary operator and interpret it as a rotation about the x -axis on a Bloch sphere (where the bottom and top states are the $|0\rangle$ and $|1\rangle$ states).

Commutators and uncertainty

The commutator between two operators is defined as $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$.

- (i) Verify that for position and momentum operators

$$[\hat{x}, \hat{p}] = i\hbar \mathbb{1},$$

using their position representation, where the operators act as

$$\hat{x} |x'\rangle = x' |x'\rangle, \quad \hat{p} |x'\rangle = -i\hbar \frac{d}{dx'} |x'\rangle.$$

The identity operator in the position basis is expressed as $\mathbb{1} = \int_{-\infty}^{\infty} dx |x\rangle\langle x|$. It satisfies

$$|\psi\rangle = \mathbb{1} |\psi\rangle = \int_{-\infty}^{\infty} dx |x\rangle \langle x|\psi\rangle = \int_{-\infty}^{\infty} dx \psi(x) |x\rangle,$$

which reconstructs any state $|\psi\rangle$ from its position-space wavefunction $\psi(x)$.

- (ii) Show that for any operators $\hat{A}, \hat{B}$,

$$[\hat{A}, \hat{B}]^\dagger = -[\hat{A}^\dagger, \hat{B}^\dagger],$$

meaning that the commutator of two hermitian operators is actually anti-hermitian.

- (iii) Compute the commutators between the Pauli operators:

$$[\hat{\sigma}_x, \hat{\sigma}_y], \quad [\hat{\sigma}_y, \hat{\sigma}_z], \quad [\hat{\sigma}_z, \hat{\sigma}_x],$$

and confirm that they satisfy the cyclic relation

$$[\hat{\sigma}_i, \hat{\sigma}_j] = 2i \epsilon_{ijk} \hat{\sigma}_k.$$

- (iv) Derive the uncertainty relation

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

for two Hermitian operators $\hat{A}$ and $\hat{B}$ where we define the uncertainties as

$$\Delta A \equiv \sqrt{\langle (\hat{A} - \langle \hat{A} \rangle)^2 \rangle}, \quad \Delta B \equiv \sqrt{\langle (\hat{B} - \langle \hat{B} \rangle)^2 \rangle}.$$

Hint: Then use the Cauchy-Schwarz inequality

$$|\langle \phi | \psi \rangle|^2 \leq \langle \phi | \phi \rangle \langle \psi | \psi \rangle.$$

on appropriate state vectors $|\phi\rangle$ and $|\psi\rangle$.

Advanced commutators - practical calculation rules for commutators

(i) Show that for any three operators $\hat{A}, \hat{B}, \hat{C}$,

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}].$$

(ii) Functions of position and momentum

(a) Use the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar \mathbb{1}$, show that

$$[f(\hat{x}), \hat{p}] = i\hbar f'(\hat{x}),$$

(b) Deduce the special case

$$[\hat{x}^n, \hat{p}] = i\hbar n \hat{x}^{n-1},$$

Remark: $[\hat{A}, \hat{B}^n] = n\hat{B}^{n-1}[\hat{A}, \hat{B}]$ in general for $[\hat{A}, \hat{B}] = c \mathbb{1}$.

(iii) Baker–Campbell–Hausdorff (BCH) – If $[\hat{A}, \hat{B}] = c \mathbb{1}$ with c a complex number, show that

$$e^{\hat{A}}e^{\hat{B}} = e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A}, \hat{B}]}$$